

# **A Purely Mathematical Analysis of a Triple-Congruence Integer System and Its Stability Properties**

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## **1. Introduction**

The purpose of this paper is to present a fully self-contained mathematical analysis of a particular triple-congruence system and the integer structure that arises from it. The focus is strictly on number theory and abstract topological invariants, with no reliance on geometric constructions, physical interpretations, or applied motivations. The goal is to formalize the congruence relations, establish their solvability, and describe the resulting integer lattice in a rigorous and purely arithmetic manner.

### **1.1 Purpose of the Mathematical Layer**

#### **1.1.1 Objective of the Study**

This work develops the mathematical framework underlying a system of three simultaneous congruences. The analysis includes the definition of the congruence system, the application of the Chinese Remainder Theorem, and the characterization of the resulting solution class as an infinite arithmetic progression.

#### **1.1.2 Exclusion of Non-Mathematical Assumptions**

All results are derived solely from classical number theory and abstract properties of integer-valued topological invariants. No geometric, physical, or interpretive assumptions are introduced at any stage.

### **1.2 Scope and Limitations**

#### **1.2.1 Scope of the Paper**

The scope of this paper is intentionally narrow: it addresses only the arithmetic structure of the congruence system and the formal topological interpretation of integer invariants. The analysis is limited to the mathematical properties that can be established without invoking external frameworks.

#### **1.2.2 Intentional Exclusion of Motivational Context**

Although congruence systems often arise from geometric or applied considerations, such motivations are deliberately omitted here. The paper restricts itself to the internal logic of the mathematical structures involved.

## **2. Preliminaries and Notation**

This section introduces the number-theoretic and topological notation used throughout the paper. All definitions are standard and self-contained, and no geometric or physical interpretations are invoked.

### **2.1 Basic Number-Theoretic Notation**

### 2.1.1 Congruence Notation

We write

$$a \equiv b \pmod{m}$$

to indicate that the integers  $a$  and  $b$  differ by a multiple of the integer  $m$ ; that is,

$$m \mid (a - b).$$

This notation is used to express equivalence classes in the ring  $\mathbb{Z}/m\mathbb{Z}$ .

### 2.1.2 Least Common Multiple

For positive integers  $m_1, m_2, \dots$ , the **least common multiple**, denoted

$$\text{lcm}(m_1, m_2, \dots),$$

is the smallest positive integer divisible by each  $m_i$ . It is used to determine the modulus of the unique solution class of a compatible system of congruences.

### 2.1.3 Arithmetic Progressions

An **arithmetic progression** is a sequence of integers of the form

$$a + dk, k \in \mathbb{Z},$$

where  $a$  is the initial term and  $d$  is the common difference. Such progressions naturally arise as solution sets of linear congruence systems.

## 2.2 Basic Topological Notation

This subsection introduces the abstract topological concepts required later in the paper. The presentation is strictly formal: no geometric intuition, physical interpretation, or applied context is included. Only the minimal definitions necessary for discussing integer-valued invariants are provided.

### 2.2.1 Abstract U(1) Principal Bundles

#### 2.2.1.1 Definition of a Principal Bundle

A **principal bundle** with structure group  $G$  consists of a total space  $P$ , a base space  $B$ , a continuous surjection

$$\pi: P \rightarrow B,$$

and a free, continuous right action of  $G$  on  $P$ , such that each point of  $B$  has a neighborhood over which the bundle is locally isomorphic to  $U \times G$ . This definition is purely topological and does not rely on any geometric structures such as metrics or connections.

### 2.2.1.2 Specialization to the Group U(1)

When the structure group is the circle group

$$U(1) = \{z \in \mathbb{C} \mid |z| = 1\},$$

the resulting object is called a **U(1) principal bundle**. Such bundles are classified, up to isomorphism, by an integer invariant when the base space is  $S^2$ .

### 2.2.1.3 Local Triviality Requirement

For every point  $b \in B$ , there exists an open neighborhood  $U \subseteq B$  such that

$$\pi^{-1}(U) \cong U \times U(1)$$

as principal U(1) bundles. This property ensures that the bundle is locally simple even if its global structure is nontrivial.

## 2.2.2 First Chern Number as an Integer

### 2.2.2.1 Abstract Definition of the First Chern Class

For a U(1) principal bundle  $P \rightarrow B$ , the **first Chern class**

$$c_1(P) \in H^2(B; \mathbb{Z})$$

is the cohomology class that classifies the bundle up to isomorphism. When the base space is  $S^2$ , the second cohomology group is

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

so the first Chern class corresponds to a single integer.

### 2.2.2.2 Integer-Valued Invariant

The integer associated with  $c_1(P)$  is called the **first Chern number**, and we write

$$c_1(P) \in \mathbb{Z}.$$

This integer uniquely determines the isomorphism class of the U(1) principal bundle over  $S^2$ .

### 2.2.2.3 Additivity Under Tensor Products

If  $P_1$  and  $P_2$  are U(1) principal bundles over the same base, then the tensor product bundle satisfies

$$c_1(P_1 \otimes P_2) = c_1(P_1) + c_1(P_2).$$

This property reflects the group structure of the classification.

### 3. The Dimensional-Closure Congruence System

This section introduces the triple-congruence system that forms the basis of the subsequent number-theoretic analysis. The presentation is entirely arithmetic: no geometric, physical, or interpretive context is invoked.

#### 3.1 Definition of the Triple Congruence System

The system under consideration consists of three simultaneous congruences imposed on an integer  $n$ . Each congruence restricts  $n$  modulo a different positive integer.

##### 3.1.1 First Congruence

The first condition requires that

$$n \equiv 1(\text{mod}5).$$

This specifies that  $n$  lies in the residue class of 1 modulo 5.

##### 3.1.2 Second Congruence

The second condition requires that

$$n \equiv -1(\text{mod}15).$$

Equivalently,

$$n \equiv 14(\text{mod}15).$$

##### 3.1.3 Third Congruence

The third condition requires that

$$n \equiv 1(\text{mod}20).$$

This places  $n$  in the residue class of 1 modulo 20.

#### 3.2 Definition of the Dimensional-Closure Set $C_{\text{dim}}$

##### 3.2.1 Formal Set Definition

The set of all integers satisfying the three congruences simultaneously is defined as

$$C_{\text{dim}} = \{ n \in \mathbb{Z} \mid n \equiv 1(\text{mod}5), n \equiv -1(\text{mod}15), n \equiv 1(\text{mod}20) \}.$$

This set is nonempty and will later be shown to form an infinite arithmetic progression.

##### 3.2.2 Basic Properties of the Set

Because the system admits a unique solution modulo the least common multiple of the moduli, the set  $C_{\text{dim}}$  is closed under addition of multiples of

$$\text{lcm}(5,15,20) = 300.$$

Thus, if  $n \in C_{\text{dim}}$ , then

#### **4. Solvability and Structure of the Congruence System**

This section establishes the solvability of the triple-congruence system introduced earlier and characterizes its complete solution set. The analysis relies solely on classical results from elementary number theory, particularly the Chinese Remainder Theorem. No external assumptions or interpretive frameworks are required.

##### **4.1 Compatibility of the Moduli**

###### **4.1.1 Pairwise Compatibility**

To determine whether the system of congruences admits a simultaneous solution, it is necessary to verify that the moduli involved are pairwise compatible. Two congruences

$$n \equiv a \pmod{m}, n \equiv b \pmod{m'}$$

are compatible if and only if

$$a \equiv b \pmod{\text{gcd}(m, m')}.$$

Applying this criterion to the moduli 5, 15, and 20 shows that the corresponding residue conditions do not conflict. Each pair of congruences satisfies the compatibility requirement, ensuring that the full system is solvable.

###### **4.1.2 Least Common Multiple**

Once compatibility is established, the modulus of the unique solution class is given by the least common multiple of the three moduli. A direct computation yields

$$\text{lcm}(5,15,20) = 300.$$

This value determines the periodicity of the complete solution set: any two solutions differ by a multiple of 300.

#### **4.2 Unique Solution Class via the Chinese Remainder Theorem**

##### **4.2.1 CRT Construction**

The Chinese Remainder Theorem guarantees that a compatible system of congruences has a unique solution modulo the least common multiple of the moduli. Applying the constructive form of the theorem, one solves the congruences sequentially or through simultaneous modular reconstruction. Either approach yields a single residue class modulo 300 that satisfies all three conditions.

#### **4.2.2 Final Result**

Carrying out the explicit computation produces the solution

$$n \equiv 137 \pmod{300}.$$

This congruence describes the entire set of integers that satisfy the original triple-congruence system.

### **4.3 Verification of the Solution**

#### **4.3.1 Check Modulo 5**

Substituting  $n = 137$  into the first congruence gives

$$137 \equiv 1 \pmod{5},$$

since  $137 - 1 = 136$  is divisible by 5.

#### **4.3.2 Check Modulo 15**

For the second congruence,

$$137 \equiv -1 \pmod{15},$$

because  $137 + 1 = 138$  is divisible by 15.

#### **4.3.3 Check Modulo 20**

For the third congruence,

$$137 \equiv 1 \pmod{20},$$

as  $137 - 1 = 136$  is divisible by 20.

These verifications confirm that 137 satisfies all three congruences simultaneously and therefore represents the unique solution class modulo 300.

## **5. Integer Lattice Structure of $C_{\dim}$**

The purpose of this section is to describe the full integer structure of the solution set



$$C_{\text{dim}} = \{n \in \mathbb{Z} \mid n \equiv 1(\text{mod}5), n \equiv -1(\text{mod}15), n \equiv 1(\text{mod}20)\}.$$

Having established in the previous section that the system admits a unique solution modulo 300, we now characterize the resulting arithmetic progression, identify its minimal positive representative, and determine its minimal prime element. All arguments rely solely on elementary number theory.

## 5.1 Arithmetic Progression Representation

### 5.1.1 General Form

Since the system of congruences has a unique solution modulo

$$\text{lcm}(5,15,20) = 300,$$

the complete solution set is given by the arithmetic progression

$$C_{\text{dim}} = \{137 + 300k \mid k \in \mathbb{Z}\}.$$

This expression captures every integer that satisfies the three congruences simultaneously.

### 5.1.2 Proof of Completeness

To show that all solutions have this form, let  $n$  be any integer satisfying the congruence system. By the Chinese Remainder Theorem, any two solutions must be congruent modulo 300. Since 137 is one such solution, it follows that

$$n \equiv 137(\text{mod}300),$$

and therefore  $n = 137 + 300k$  for some integer  $k$ . Conversely, substituting  $n = 137 + 300k$  into each congruence shows that all such integers satisfy the system. This establishes completeness.

## 5.2 Minimal Positive Representative

### 5.2.1 Proof of Minimality

The integer 137 is the smallest positive element of the progression

$$137 + 300k.$$

If  $k \geq 0$ , the smallest value occurs at  $k = 0$ , yielding 137. If  $k < 0$ , then

$$137 + 300k < 0,$$

so no negative value of  $k$  produces a positive integer smaller than 137. Thus, 137 is the minimal positive representative.

### 5.2.2 Exclusion of Smaller Candidates

To confirm minimality directly, one may check that no integer  $1 \leq n < 137$  satisfies all three congruences. Since the solution class is unique modulo 300, any smaller integer would have to be congruent to 137 modulo 300, which is impossible for a positive integer less than 137. Therefore, no smaller candidate exists.

## 5.3 Minimal Prime Representative

### 5.3.1 Primality of 137

The integer 137 is prime. A direct verification shows that it has no positive divisors other than 1 and itself. In particular, it is not divisible by any prime less than or equal to  $\sqrt{137} < 12$ , namely 2, 3, 5, 7, or 11.

### 5.3.2 No Smaller Prime in the Progression

To determine whether the progression contains a smaller prime, consider any integer of the form

$$137 + 300k$$

with  $k < 0$ . For  $k = -1$ , the value is  $-163$ , which is negative; for all smaller  $k$ , the values remain negative. Since no positive integer smaller than 137 lies in the progression, and since 137 itself is prime, it follows that 137 is the smallest prime element of  $C_{\text{dim}}$ .

## 6. Abstract Topological Interpretation

This section provides a purely formal topological interpretation of the integer structure described earlier. The discussion is restricted to classical results concerning  $U(1)$  principal bundles over the 2-sphere and the integer invariants that classify them. No geometric intuition, physical interpretation, or application-specific context is introduced.

### 6.1 Classification of $U(1)$ Bundles over $S^2$

#### 6.1.1 Classical Classification Theorem

A fundamental result in the theory of principal bundles states that  $U(1)$  principal bundles over the 2-sphere  $S^2$  are classified, up to isomorphism, by the second cohomology group

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}.$$

This means that each isomorphism class corresponds uniquely to an integer, and conversely, every integer arises from some  $U(1)$  principal bundle over  $S^2$ .

### 6.1.2 Integer Invariant Interpretation

Under this classification, the integer associated with a  $U(1)$  principal bundle serves as a complete invariant: two bundles are isomorphic if and only if they determine the same integer. Thus, the set of all  $U(1)$  principal bundles over  $S^2$  can be identified with the set of integers, and the classification is both exhaustive and non-redundant.

## 6.2 First Chern Number as an Integer Invariant

### 6.2.1 Integer-Valued Nature

For a  $U(1)$  principal bundle  $P \rightarrow S^2$ , the first Chern class

$$c_1(P) \in H^2(S^2; \mathbb{Z})$$

corresponds to a unique integer under the isomorphism

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}.$$

This integer is called the **first Chern number** of the bundle. It provides a complete and intrinsic characterization of the bundle's isomorphism class.

### 6.2.2 Additivity Under Tensor Products

If  $P_1$  and  $P_2$  are  $U(1)$  principal bundles over  $S^2$ , then their tensor product satisfies the relation

$$c_1(P_1 \otimes P_2) = c_1(P_1) + c_1(P_2).$$

This additivity reflects the group structure of the classification and plays a central role in understanding how integer invariants behave under natural bundle operations.

## 6.3 Compatibility with Congruence Classes

### 6.3.1 If $c_1(P) \in \mathcal{C}_{\dim}$

If the first Chern number of a  $U(1)$  principal bundle lies in the set

$$C_{\text{dim}} = \{137 + 300k \mid k \in \mathbb{Z}\},$$

then it must take the form

$$c_1(P) = 137 + 300k$$

for some integer  $k$ . This follows directly from the arithmetic characterization of the congruence system and the fact that the Chern number is an integer.

### 6.3.2 Completeness of the Characterization

Conversely, any integer of the form  $137 + 300k$  may serve as the first Chern number of a  $U(1)$  principal bundle over  $S^2$ . Since every integer corresponds to a unique isomorphism class of such bundles, the congruence class fully determines the set of admissible topological invariants. Thus, the correspondence between  $C_{\text{dim}}$  and the possible Chern numbers is complete.

## 7. Mathematical Stability of the Structure

This section examines the stability properties of the solution set

$$C_{\text{dim}} = \{137 + 300k \mid k \in \mathbb{Z}\},$$

focusing on its behavior under integer shifts, the rigidity of its congruence class, and the independence of its derivation from any geometric or external assumptions. All arguments rely exclusively on classical number theory.

### 7.1 Stability Under Integer Shifts

#### 7.1.1 Closure Under Addition

The set  $C_{\text{dim}}$  is closed under addition of integer multiples of 300. If  $n \in C_{\text{dim}}$ , then by definition

$$n \equiv 137 \pmod{300}.$$

For any integer  $k$ , the shifted value

$$n + 300k$$

satisfies

$$n + 300k \equiv 137 + 300k \equiv 137 \pmod{300},$$

and therefore remains in  $C_{\text{dim}}$ . This demonstrates that the structure of the solution set is stable under all integer shifts by the modulus of the congruence system.

### 7.1.2 Lattice Interpretation

The closure property implies that  $C_{\text{dim}}$  forms a one-dimensional integer lattice embedded in  $\mathbb{Z}$ , generated by the base point 137 and the step size 300. Formally,

$$C_{\text{dim}} = 137 + 300\mathbb{Z}.$$

This representation highlights the regularity and periodicity of the solution set and provides a natural algebraic structure for analyzing its stability.

## 7.2 Structural Rigidity of the Congruence Class

### 7.2.1 Uniqueness Modulo 300

The Chinese Remainder Theorem guarantees that the triple-congruence system admits exactly one solution class modulo 300. Thus, if  $n$  and  $m$  are both solutions, then

$$n \equiv m \pmod{300}.$$

This uniqueness ensures that the arithmetic progression describing  $C_{\text{dim}}$  is the only possible progression compatible with the congruence system.

### 7.2.2 No Alternative Classes

To establish rigidity, suppose there exists another integer  $r$  satisfying the same congruences but not congruent to 137 modulo 300. This would contradict the uniqueness guaranteed by the Chinese Remainder Theorem. Therefore, no alternative congruence class exists, and the structure of  $C_{\text{dim}}$  is completely determined by the modulus 300 and the representative 137.

## 7.3 Independence from Geometric Assumptions

### 7.3.1 Purely Arithmetic Derivation

All results concerning the solvability, structure, and stability of the congruence system follow directly from classical number theory. The arguments rely solely on divisibility properties, modular arithmetic, and the Chinese Remainder Theorem. No geometric constructions, interpretive frameworks, or external assumptions are required.

### 7.3.2 No External Assumptions Required

The integer lattice structure

$$137 + 300\mathbb{Z}$$

is a consequence of the arithmetic compatibility of the moduli and the uniqueness of the solution class. Its stability and rigidity arise entirely from these intrinsic properties, independent of any additional context.

## **8. Discussion and Future Mathematical Directions**

The preceding sections establish the solvability, structure, and stability of a specific triple-congruence system and its associated integer lattice. Although the analysis is complete within its intended scope, the mathematical framework naturally suggests several avenues for further investigation. These directions remain entirely within number theory and abstract topology, without invoking any geometric or physical interpretations.

### **8.1 Open Number-Theoretic Questions**

#### **8.1.1 Generalization to Other Moduli Triples**

The congruence system studied in this paper involves the moduli 5, 15, and 20. A natural extension is to consider other triples of moduli and to classify the conditions under which such systems admit unique solutions modulo their least common multiple. This includes identifying compatibility criteria, determining the structure of the resulting arithmetic progressions, and analyzing how the choice of residues influences the solution class. Such generalizations may reveal broader patterns in the behavior of multi-modulus congruence systems.

#### **8.1.2 Structure of Multi-Modulus Systems**

Beyond triples, one may investigate congruence systems involving four or more moduli. The Chinese Remainder Theorem extends naturally to such settings, but the combinatorial and algebraic structure becomes increasingly intricate. Questions of interest include the characterization of solvability conditions, the classification of solution lattices, and the identification of minimal representatives within higher-dimensional congruence frameworks. These investigations may contribute to a deeper understanding of modular arithmetic in complex settings.

### **8.2 Abstract Topological Extensions**

### 8.2.1 Higher-Dimensional Integer Invariants

The classification of  $U(1)$  principal bundles over  $S^2$  by integer invariants suggests a broader inquiry into integer-valued topological invariants in higher dimensions. For example, one may study characteristic classes associated with bundles over higher-dimensional spheres or other manifolds, focusing on the algebraic properties of the corresponding cohomology groups. Such investigations remain purely formal and do not require geometric interpretation.

### 8.2.2 Other Compact Lie Groups

Another direction involves extending the classification framework to principal bundles with structure groups beyond  $U(1)$ , such as  $U(n)$ ,  $SU(2)$ , or other compact Lie groups. These groups possess richer cohomological structures, and their associated characteristic classes may exhibit more complex integer or torsion behavior. Exploring these invariants within a purely algebraic and topological context may yield insights into the general theory of bundle classification.

## 9. Conclusion

This paper has presented a complete and self-contained mathematical analysis of a specific triple-congruence system and the integer lattice that arises from it. Beginning with the formal definition of the congruences, we established their compatibility, applied the Chinese Remainder Theorem to determine the unique solution class modulo 300, and demonstrated that the resulting set of solutions forms the arithmetic progression

$$137 + 300\mathbb{Z}.$$

The minimal positive representative and minimal prime representative were identified through direct number-theoretic arguments, and the structure of the solution set was shown to possess both stability under integer shifts and rigidity with respect to its defining congruence class.

In addition to the arithmetic analysis, the paper provided a purely formal topological interpretation by recalling the classical classification of  $U(1)$  principal bundles over  $S^2$  and the integer-valued nature of their first Chern numbers. This interpretation remained strictly abstract and served only to illustrate how integer invariants naturally align with the arithmetic structure of the congruence system, without invoking any geometric or physical context.

All results were derived exclusively from elementary number theory and abstract topology. No external assumptions were introduced, and the analysis remained entirely

within the mathematical layer. The framework developed here demonstrates how a simple system of congruences can give rise to a well-defined and stable integer lattice, while also suggesting several avenues for further exploration in both number theory and the study of integer-valued topological invariants.

## **10. Safety Appendix**

This appendix enumerates the categories of content that must not appear anywhere in the paper. These restrictions ensure that the work remains strictly within the domain of pure mathematics and abstract topology, without introducing geometric, physical, engineering, or system-specific interpretations. Each prohibited category is listed with explicit sub-items to prevent accidental overreach.

### **10.1 Forbidden: Geometric Motivation**

#### **10.1.1 No Projections**

The paper must not reference projection maps, dimensional reductions, or any geometric transformations.

#### **10.1.2 No Embeddings**

Embeddings of manifolds, spaces, or structures must not be mentioned in any context.

#### **10.1.3 No Curvature**

Curvature, curvature forms, or curvature-related quantities must not appear, even in abstract form.

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Holonomy groups, Kähler manifolds, or any related geometric structures must not be referenced.

### **10.2 Forbidden: Physical Interpretation**

#### **10.2.1 No Coupling Constants**

The paper must not mention or allude to any physical coupling constants.

#### **10.2.2 No Fine-Structure Constant**



No reference may be made to the fine-structure constant or any physical constant that could be associated with the integer 137.

### **10.2.3 No Render Layer or Subject Layer**

Layered physical or metaphysical models must not be included in any form.

## **10.3 Forbidden: Engineering Pathways**

### **10.3.1 No Chip Architecture**

The paper must not describe or imply any computational or hardware architecture.

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No mention may be made of processing units, compute modules, or engineering constructs.

### **10.3.3 No Implementation Details**

Algorithms, design pathways, or steps that could be interpreted as implementation guidance must not be included.

## **10.4 Forbidden: Internal Structural Frameworks**

### **10.4.1 No Dimensional-Closure Axiom as Physics**

The congruence system must not be presented as a physical law, principle, or axiom.

### **10.4.2 No Structural Cycles**

Structural cycles, feedback loops, or system-level dynamics must not be referenced.

### **10.4.3 No Integrability Conditions**

Integrability assumptions, constraints, or conditions must not be introduced, even abstractly.

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